

Interview Questions

1. How does a decision tree work?

A decision tree works by splitting the data into subsets based on the value of input features. At each node, it chooses the feature that best separates the classes using a criterion like Gini impurity or entropy. This continues recursively, forming a tree structure where leaves represent final class predictions.

2. What is entropy and information gain?

Entropy measures the impurity or randomness in the data. Information Gain is the reduction in entropy after splitting the data. In decision trees, we choose the feature that gives the highest information gain to make the best split.

3. How is random forest better than a single tree?

A random forest combines multiple decision trees to improve performance. Each tree is trained on a random subset of data and features. The final prediction is made by majority voting. This reduces overfitting and improves generalization.

4. What is overfitting and how do you prevent it?

Overfitting happens when a model learns the noise in training data instead of the actual pattern. In decision trees, it can be prevented by limiting the depth of the tree, setting a minimum number of samples per leaf, or using ensemble methods like Random Forest.

5. What is bagging?

Bagging, or Bootstrap Aggregating, is an ensemble technique that trains multiple models on different random samples of the data. The final output is the average (for regression) or majority vote (for classification). Random Forest is a bagging-based model.

6. How do you visualize a decision tree?

In Python, we can use `plot_tree` from `scikit-learn`'s `tree` module or export it to `Graphviz`. This allows us to see the structure of the tree including splits, thresholds, and predicted outcomes.

7. How do you interpret feature importance?

In tree-based models, feature importance tells us how much each feature contributes to the model's predictions. It is calculated based on the reduction in impurity from splits that involve that feature. Higher importance means more influence.

8. What are the pros/cons of random forests?

Pros: High accuracy, robust to overfitting, works well on many problems, handles missing data.

Cons: Less interpretable than single trees, slower to predict, and may require more memory.